Daevik Jain

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Technical Skills

Languages: Python, TypeScript, JavaScript, C++, C, Java, SQL, Bash, HTML/CSS, MATLAB
Frameworks & Libraries: React, Next.js, FastAPI, PyTorch, TensorFlow, Playwright, OpenCV, scikit-learn, Pandas
Tools & Technologies: AWS, Terraform, Docker, PostgreSQL, Redis, Supabase, Node.js, REST APIs, Git, Linux, Jira

Experience

QA Engineer Jan 2026 – Apr 2026
Relay Financial *Toronto, ON*

- Engineered an AWS Lambda test account generator with DynamoDB and S3, saving **$25K** in annual testing costs.
- Developed a React/Vue.js/TypeScript platform powering bug bashes and release workflows for **80+ engineers**.
- Built GitHub Actions and Jira integrations to automate test tracking and bug reporting, saving **5+ hours/week**.
- Maintained **150+ Playwright** tests at a **96% pass rate**, catching **50+ critical bugs** pre-production.

Software Engineer in Test May 2025 – Aug 2025
RideCo On-Demand Transit *Waterloo, ON*

- Developed Python/Django features for a production AI transit agent across behaviour, routing, and admin configuration.
- Reduced LLM costs by **70%** via dynamic model routing, prompt token optimization, and context window tuning.
- Built an LLM eval framework with simulated user agents, LLM judge, and tool call judge to measure agent accuracy.

Machine Learning Engineer Sep 2024 – Apr 2025
Biomechatronic Design Team *Waterloo, ON*

- Built a TensorFlow/Pandas pipeline classifying 8 hand gestures from EMG data, achieving **91.56% test accuracy**.
- Applied wavelet transforms to denoise EMG signals, reducing misclassification between similar gestures.
- Developed a Matplotlib + Tkinter tool for real-time EMG signal visualization, used by the team for debugging.

Projects

Avesia – Winner at NexHacks | ***React, FastAPI, Python, MongoDB, OpenCV, Tailwind***

- Built a no-code platform enabling non-technical users to compose and deploy real-time AI surveillance pipelines.
- Architected a full-stack React + FastAPI + MongoDB app with a Gemini-powered vision agent and REST API.
- Implemented OpenCV motion detection to gate API calls, reducing unnecessary inference requests.

Compass – Winner at QHacks | ***FastAPI, pgvector, Supabase, PostgreSQL, React, scikit-learn***

- Built an AI-powered civic dashboard to query 137k+ service requests and generate data-backed analytical reports.
- Designed a RAG pipeline with FastAPI, Supabase, and pgvector, integrating a Gemini agent for trend analysis.
- Implemented UMAP to project 768-dimensional vector embeddings into an interactive 3D cluster visualization.

Blurr – Semifinalist at HackTheNorth | ***FastAPI, OpenCV, WebRTC, React, Python, TypeScript***

- Created a real-time computer vision privacy shield that detects and blurs sensitive information in live video streams.
- Built a low-latency YOLOv8, OpenCV, and WebRTC pipeline achieving **sub-100ms** in-memory video redaction.

Solari – HackThe6ix | ***Python, C, Embedded Systems, QNX, TwelveLabs***

- Developed an AI assistive wearable with real-time audio and haptic feedback for the visually impaired.
- Built a multimodal pipeline with TwelveLabs and Gemini APIs for real-time scene analysis and narration.
- Architected a dual-Raspberry-Pi system isolating QNX for sensor processing and Linux for AI inference.

AI Platformer | ***PyTorch, C++, SFML***

- Built a 2D platformer with a native PyTorch C++ reinforcement learning agent to autonomously beat parkour levels.
- Implemented a dual-head policy network with sigmoid jump and softmax movement outputs.
- Designed a reward-normalized policy gradient loop with Xavier initialization and a sliding experience replay buffer.

Education

University of Waterloo Sep. 2024 – Present
BASc in Biomedical Eng, Option in Software Eng, President Scholarship of Distinction, 3.9 GPA *Waterloo, ON*